



Programming Fundamentals

Class Assignment # 4

Total Marks: 20

Deadline: 24-06-25: till 11:59 PM

Instructions:

- 1) You have to upload your assignments on CMS.
 - CPP should be combined with files into zip folder.
- 2) All program codes should be written in C/C++. Students should use Visual studio compiler for coding.
- 3) Plagiarism in the work is Professional dishonesty which leads to strict disciplinary action and zero in grading of Assignment.
- 4) **Make your own functions and every function should perform one functionality only.**
- 5) **Use pointer notation rather than array notation.**
- 6) **Don't use of any built-in function related to char array otherwise your task will be cancelled.**
- 7) **Don't use Global variables and string data type.**

Topics: Demonstrate knowledge of Pointers, **2D dynamic memory allocation, function, file handling in C++ in programming (C++).**

CLO NO	CLO STATEMENT	PLO	Taxonomy LEVEL
3	Be able to use functions/modular programming with elementary constructs of a programming language to solve complex problems.	2	C3

Grading Rubric

Parameters	Weights	Remarks
Completeness	50%	Will determine by checking whether your code implements the tasks we ask to implement
Logic	10%	Logic building
Correctness	20%	Incorrect code behavior or not implementing something you are being told to.
Coding Standards	20%	Your code contains comments, well indented and easy to read. Make use of descriptive variable names and user prompts. For function-based programs the code should be organized in as many functions as required and each function contains single task.

Question 1: Seat Assignment for Commercial Airplane Using Jagged Array [10-Points]

Write a C++ program that assigns seats for a commercial airplane using a jagged array. The airplane has 13 rows, with a varying number of seats in each row. The seats are divided into three classes:

- **First Class:** Rows 1 and 2, each row may have a different number of seats (e.g., 4 seats in Row 1, 6 seats in Row 2).
 - **Business Class:** Rows 3 through 7, with different numbers of seats (e.g., 6 seats in Row 3, 4 seats in Row 4).
 - **Economy Class:** Rows 8 through 13, again with varying seat numbers.
1. **Read seat availability from a file:** The file should provide the number of seats available in each row for each class, as well as the status (whether the seat is available or not).
 2. **Prompt the user for ticket type:** The program should prompt the user to select a ticket type (First Class, Business Class, or Economy Class).
 3. **Prompt the user for desired seat:** Based on the chosen ticket type, ask the user to select a specific seat by indicating the row and seat number. The program should verify whether the seat is available, and if so, assign it.
 4. **Find the row and column:** The program should use a function to find the row and column from the file data, ensuring that the seat chosen by the user corresponds to a valid seat in the specified class.
 5. **Update the seat availability:** Once a seat is assigned, the program should update the file to reflect that the seat is no longer available.
 6. **Display the updated seat map:** After assigning a seat, display the updated seating arrangement showing which seats are available and which are taken.

Map.txt:

	A	B	C	D	E	F
Row 1	*	*	X	*	-	-
Row 2	*	X	*	X	-	-
Row 3	*	*	X	X	*	-
Row 4	X	*	X	*	X	-
Row 5	*	X	*	X	*	-
Row 6	*	X	*	*	*	-
Row 7	X	*	*	*	X	-
Row 8	*	X	*	X	X	*
Row 9	X	*	X	X	*	X
Row 10	*	X	*	X	X	X
Row 11	*	*	X	*	X	*
Row 12	*	*	X	X	*	X
Row 13	*	*	*	*	X	*

Here, * indicates that the seat is available; X indicates that the seat is occupied. Make this a menu-driven program; show the user's choices and allow the user to make the appropriate choices. - means Not part of the seating arrangement (indicating an invalid or unavailable seat for that row).

Question 2:**Using divide and Conquer approach****[10-Points]**

A company hired 8 temporary workers who are paid hourly and you are given a data file that contains the last name of the employees, the number of hours each employee worked in a week, and the hourly pay rate of each employee. You are asked to write a program that computes each employee's weekly pay and the average salary of all the workers.

The program then outputs the weekly pay of each employee, the average weekly pay, and the names of all the employees whose pay is greater than or equal to the average pay. If the number of hours worked in a week is more than 40, then the pay rate for the hours over 40 is 1.5 times the regular hourly rate. Use two parallel arrays: a two-dimensional array to store the names of all the employees, and a two-dimensional array of 10 rows and 3 columns to store the number of hours an employee worked in a week, the hourly pay rate, and the weekly pay.

Your program must contain at least the following functions:

- A function to read the data from the file into the arrays,
- A function to determine the weekly pay.
- A function to output the names of all the employees whose pay is greater than or equal to the average weekly pay,
- A function to output each employee's data.

Expected Output

Employee Last Name Hours Worked Hourly Pay Rate			

Ali	35	15.00	+-----+-----+-----+-----+
Khan	42	18.00	Employee Last Hours Worked Hourly Pay Rate Weekly Pay (\$)
Ahmed	38	16.50	Name
Hassan	40	17.00	+-----+-----+-----+-----+
Mahmood	45	16.00	Ali 35 \$15.00 \$525.00
Abbas	37	14.50	Khan 42 \$18.00 \$756.00
Rahman	50	20.00	Ahmed 38 \$16.50 \$627.00
Hussain	39	15.00	Hassan 40 \$17.00 \$680.00
			Mahmood 45 \$16.00 \$720.00
			Abbas 37 \$14.50 \$536.50
			Rehman 50 \$20.00 \$1000.00
			Hussain 39 \$15.00 \$585.00
			+-----+-----+-----+-----+
Average Weekly Pay: \$683.69			
Employees with Weekly Pay Greater Than Average:			

Khan			
Mahmood			
Rehman			